RAJALAKSHMI ENGINEERING COLLEGE

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C2

Started on	Thursday, 17 September 2026, 8:18 AM
State	Finished
Completed on	Thursday, 17 September 2026, 8:21 AM
Time taken	2 mins 24 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
   int rev = 0, remainder;
   while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;

    }
print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main() {
    int n, rev = 0, remainder;
    int count = 0; // Counter variable to track operations

    // Read input silently without any printf prompt strings
    if (scanf("%d", &n) != 1) return 0;

    // --- Counter Method Implementation ---

    rev = 0;
    count++; // For: rev = 0;

    count++; // For the initial condition check of the while loop (n != 0)
    while (n != 0) {
        remainder = n % 10;
        count++; // For: remainder = n % 10;

        rev = rev * 10 + remainder;
        count++; // For: rev = rev * 10 + remainder;

        n /= 10;
        count++; // For: n /= 10;

        count++; // For the subsequent condition check of the while loop
    }

    count++; // For: print(rev); (Conceptual print operation)

    // Output only the final value of the counter variable
    printf("%d\n", count);

    return 0;
}
```

```
37
```

	Input	Expected	Got	
✔	12	11	11	✔
✔	1234	19	19	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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